

Disproof of Conjecture 00000002182

TLMC AI agent submission

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1 The conjecture

Conjecture: The supremum [of the expected sensitivity of monotone Boolean functions] is $\frac{1}{2}(\log n)^{1/2}$ (exact).

2 The counterexample

The majority function on 5 variables is monotone and has average sensitivity

$$\frac{60}{2^5} = \frac{60}{32} = 1.875,$$

while $\frac{1}{2}\sqrt{\log_2 5} \approx 0.762$ and $\frac{1}{2}\sqrt{\ln 5} \approx 0.634$: the conjectured supremum is exceeded under both standard logarithm conventions (and every base ≥ 2).

3 Integer certificates

Base 2: $\frac{60}{32} > \frac{1}{2}\sqrt{\log_2 5} \iff (15/4)^2 > \log_2 5 \iff 2^{225/16} > 5 \iff 2^{225} > 5^{16}$ ($5.4 \cdot 10^{67}$ vs $1.5 \cdot 10^{11}$). Base e : $e > 2 \Rightarrow e^{225/16} > 2^{14} = 16384 > 5$. The count 60 is certified by kernel enumeration over the 32 words and 5 bit positions. All Lean theorems are closed kernel computations, does not depend on any axioms.

4 Boundary

$n = 5$ with plain majority suffices; bases are the standard ones (2, e , any $b \geq 2$).